Causal Histories, Observable States, and Cubical Geometry

Alexander Kolpakov
Center for Science, Technology, Engineering, and Mathematics
University of Austin, Austin, Texas, United States
akolpakov@uaustin.org
The Wolfram Institute for Computational Foundations of Science
akolpakov@wolframinstitute.org

Research note, 20 October 2026

Abstract

Finite hypergraphs and token-replacement histories have explicit presentations by restrictions of permutation torsors. Their symmetry quotients retain the usual inverse-automorphism weights; forgetting the order of events introduces an additional linear-extension factor. For a fixed occurrence history, prefixes form the familiar CAT(0) cube complex of ideals of a causal poset. We give a direct criterion for an observation of those prefixes to inherit their median operation: its fibres must be obtained by forgetting a fixed subset of events. For additive token observations this becomes an exact condition on the integer effect matrix, with a finite witness whenever it fails. Finally, terminating confluent one-token systems can have state complexes with fundamental group of any positive finite free rank, although their history unfoldings are trees. The results distinguish geometry of occurrences from geometry of observable states.

1 Introduction

An occurrence history records which event produced each token, which event consumed it, and which events depended on earlier ones. An observable state may retain only the tokens currently present. These are different objects. In particular, identifying occurrences with the same observable state need not preserve their geometry.

The geometry of event structures and state complexes is established theory. Nielsen–Plotkin–Winskel [7] relate Petri nets, event structures and domains. Ghrist–Peterson [5] study cubical state complexes for reconfiguration systems. Ardila–Owen–Sullivant [1, Section 2] give a poset description of CAT(0) cube complexes. Chalopin–Chepoi [4, Sections 3 and 8] relate 1-safe Petri nets to special cube complexes and describe their unfoldings using universal covers. We use these connections as antecedents, rather than proposing the CAT(0) geometry of histories as a new theorem.

Our setting is finite and explicit. A history consists of a fixed inventory of token-replacement occurrences, with no absence tests or other global application conditions. Two preserved blocks inside one permutation torsor describe birth-to-consumption wiring and schedules. Forgetting schedules produces a linear-extension weight, which must be distinguished from the graph automorphism weight.

The main question concerns an observation $\sigma : \mathcal{J}(P) \to S$ of the ideals of a finite causal poset P. The median of three ideals is their coordinatewise majority. We prove that this operation is well defined on observable states exactly when equality of observations means equality on a fixed set of

event coordinates. We then specialize the criterion to integer token-effect matrices and give explicit witnesses to failure. This is the finite ideal-coordinate form of established median quotient theory; the proof here uses only convexity and parallel event edges. See Bowditch [3, Sections 3, 7–9] for the general median-algebra and wall framework.

A separate construction shows the limits of confluence. For any $k \geq 2$ and $L \geq 3$, a graded, terminating, confluent one-token system has a state complex with fundamental group F_{k-1} and girth $2L$. Every individual history is an interval, and the full history unfolding is a tree. Thus confluence alone does not make state identification preserve CAT(0) geometry.

The contribution is an explicit account of the restrictions, symmetry and schedule weights, together with the observation test and its token-matrix consequences. The elementary quotient argument does not constitute a new general theory of median spaces. All proofs needed for the finite statements are included; the only geometric criterion imported is the standard cubical link criterion.

2 Permutation torsors and incidence structures

2.1 A torsor over the integers

For finite sets A, B of size D, put $T(A,B) = \mathrm{Bij}(A,B)$. The actions of $\mathrm{Sym}(A)$ by precomposition and $\mathrm{Sym}(B)$ by postcomposition commute and are both free and transitive. Thus $T(A,B)$ is a bitorsor; an identification with S_D requires choosing one bijection.

There is a useful integral presentation. Introduce coordinates p_{ab}, $a \in A$, $b \in B$, and impose

$$p_{ab}^2 = p_{ab}, \qquad p_{ab}p_{ac} = 0 \quad (b \neq c), \qquad p_{ab}p_{cb} = 0 \quad (a \neq c), \tag{2.1}$$
$$\sum_b p_{ab} = 1, \qquad \sum_a p_{ab} = 1.$$

Proposition 2.1. *The resulting coordinate ring is $\prod_{\tau \in T(A,B)} \mathbb{Z}$. It represents the constant permutation torsor over $\mathbb{Z}$.*

Proof. For each bijection τ, set $e_\tau = \prod_a p_{a,\tau(a)}$. These are pairwise orthogonal idempotents. Expanding the product of the row sums gives $\sum_\tau e_\tau = 1$: every noninjective choice vanishes by column orthogonality. Moreover, $p_{ab} = \sum_{\tau(a)=b} e_\tau$. Hence the ring is generated by these components. Evaluation at each permutation identifies its component with $\mathbb{Z}$, proving the claim. Composition of bijections gives the asserted torsor actions. □

A forbidden finite pattern of wires $Q \subseteq A \times B$ is excluded by $\prod_{(a,b)\in Q} p_{ab} = 0$. Such equations cut out unions of components of the torsor. The restricted set generally has no transitive group action. Over every field the same equations select the same finite configurations. A coefficient-field change alone therefore adds no metric or approximation theorem.

Orthogonality in (2.1) is needed in positive characteristic. Over $\mathbb{F}_2$, the 4×4 matrix with zero diagonal and all other entries one is invertible and has row and column sums one, but is not a permutation matrix.

2.2 Graphs and their symmetry factors

Fix vertex degrees d_v and hyperedge arities k_e with common sum D. Partition $A = \coprod_v A_v$ and $B = \coprod_e B_e$ into blocks of those sizes. A bijection connects an incidence at v to a port of e. Repeated incidences are allowed. Ordered hyperedges retain their port order; unordered ones forget it.

Let H_V consist of permutations inside the vertex blocks and permutations of blocks of equal degree. Define H_E similarly. For coloured structures, restrict these groups to preserve colours.

Proposition 2.2. *The action groupoid of $H_V \times H_E$ on $T(A,B)$ is equivalent to the groupoid of incidence hypergraphs with the specified profiles. For any invariant restriction $X \subseteq T(A,B)$,*

$$\sum_{[G] \in X/(H_V \times H_E)} \frac{1}{|\operatorname{Aut}(G)|} = \frac{|X|}{|H_V||H_E|}. \tag{2.2}$$

Proof. Label the incidences of any such hypergraph by A_v and its ports by B_e. Its wires then give a bijection. Changing these labels is exactly the displayed action. An incidence isomorphism gives the corresponding two block permutations, and it preserves wires exactly when it stabilizes the bijection. Thus the stabilizer is the full incidence automorphism group, including permutations of repeated incidences. Orbit–stabilizer proves (2.2). □

This is the configuration model and standard groupoid cardinality [2]. Graph constraints have explicit equations. To forbid repeated incidence at (v,e), forbid every two-wire pattern in $A_v \times B_e$. Connectedness is expressed, for every nontrivial cut $U \subset V \sqcup E$, by

$$\prod_{(a,b) \text{ crossing } U} (1 - p_{ab}) = 0.$$

An empty product is one, correctly excluding an impossible cut.

For ordinary cubic multigraphs take $d_v = 3$, $k_e = 2$, and restrict to connected graphs. The pushforward of the uniform configuration measure is

$$\pi_N(G) = \frac{1}{C_N |\operatorname{Aut}(G)|}, \qquad C_N = \sum_G \frac{1}{|\operatorname{Aut}(G)|}.$$

This is the stationary law of the local Whitehead chain [6]. At two vertices, the theta and dumbbell have automorphism orders 12 and 8; their 288 and 432 configurations give probabilities 2/5 and 3/5.

3 Histories as restrictions of one permutation torsor

3.1 The replacement model

A token is a typed ordered tuple of vertex names. Equal tuples may occur as different tokens. A rule occurrence consumes a specified multiset of tokens and creates a specified multiset. Vertex names are immutable. A fresh name has a unique birth occurrence, first appears among its outputs, and is distinct from all initial names and all other fresh births.

Fix a finite set E of distinguished rule occurrences, their vertex substitutions, and initial and final token positions. The substitution for each occurrence is assumed to satisfy the rule's local equality and inequality conditions. Vertex deletion independent of token deletion and global absence conditions are not part of this model. These restrictions ensure that incomparable occurrences commute.

Let O be the disjoint union of the initial tokens and all event outputs. Let I be the disjoint union of the final token positions and all event inputs. Assume $|O| = |I| = m$ and $|E| = k$. Form the disjoint unions $A = O \sqcup \{1, \dots, k\}$ and $B = I \sqcup E$, and consider the single permutation torsor

$$T_{\text{hist}} = \operatorname{Bij}(A, B). \tag{3.1}$$

Forbid wires between the two different kinds of blocks: $p_{o,e} = p_{s,i} = 0$ for $o \in O$, $e \in E$, $s \in \{1, \dots, k\}$ and $i \in I$. The resulting restriction is canonically $\operatorname{Bij}(O, I) \times \operatorname{Bij}(\{1, \dots, k\}, E)$. Indeed, a block-preserving bijection has exactly these two restrictions, and their disjoint union is its unique inverse

construction. Write $W_{oi} = p_{o,i}$ and $R_{se} = p_{s,e}$. A point now consists of a birth-to-consumption wiring and an event order. Restrict it by the following conditions:

1. a wire joins equal token types and equal ordered vertex tuples;
2. its producer precedes its consumer, with initial time 0 and final time $k+1$;
3. every occurrence other than the birth that uses a fresh vertex follows that birth.

These are algebraic restrictions of the integral torsor. For example, if W_{oi} indicates a wire from event u to event v, and R_{su} indicates event u at position s, impose

$$W_{oi}R_{su}R_{tv} = 0 \qquad (s \geq t).$$

Incompatible endpoint types impose $W_{oi} = 0$. The vertex-birth constraints have the same form.

Theorem 3.1. *The points of this restriction correspond bijectively to valid sequential histories with the specified distinguished occurrences, tokens, ports, vertex substitutions and boundaries. For a fixed acyclic wiring, the fibre obtained by forgetting the event order is the set $\mathcal{L}(P)$ of linear extensions of its causal poset P.*

Proof. Follow the event order. Every input of the next occurrence has a unique wired producer, which is initial or has already occurred. The token has not been consumed earlier because the wiring is bijective. Its type and vertex tuple agree with the input. The fresh-name constraints also hold. The occurrence is therefore executable. Induction executes every occurrence, and the unconsumed tokens are exactly those wired to the final boundary.

Conversely, each token in a valid framed history has one birth and one consumption or final-boundary position. This defines a bijection $O \to I$ satisfying the restrictions, inverse to the construction.

The producer-before-consumer relations and the vertex-birth relations generate P. The allowed orders are precisely its linear extensions. Incomparable events consume disjoint tokens and have no fresh-name dependency. They may mention a common immutable vertex, which introduces no further constraint. Hence exchanging two adjacent incomparable events preserves execution. Every linear extension is therefore a schedule of the same occurrence history. □

The assertion concerns a fixed finite inventory. A finite budget bounds the number of occurrences, tokens and fresh names. With a fixed finite rule set and initial state, there are only finitely many inventories and vertex substitutions up to relabelling. Choose one framing of each inventory. The following construction puts their disjoint union inside one torsor without adding a multiplicity from unused labels.

Proposition 3.2. *Let $X_j \subseteq \mathrm{Bij}(A_j, B_j)$, $1 \leq j \leq q$, be finitely many restrictions given by polynomial equations in permutation coordinates. Set $d = \max_j |A_j|$. The disjoint union $\coprod_j X_j$ has an explicit presentation as a restriction of a permutation torsor on $d+q$ elements. The presentation preserves points and counts over every field.*

Proof. Choose labels on each A_j, B_j and pad its permutation to d elements by fixing all added labels. In a torsor on $d+q$ elements, require the first d and the last q labels to be preserved as blocks. Write p and s for the coordinates on these blocks. Index the second block by $\mathbb{Z}/q\mathbb{Z}$ and choose its q distinct cyclic shifts $c_j(a) = a + j - 1 \pmod q$ and set

$$u_j = \prod_{a \in \mathbb{Z}/q\mathbb{Z}} s_{a,c_j(a)}.$$

Impose $\sum_j u_j = 1$. Orthogonality of permutation coordinates makes the u_j pairwise orthogonal idempotents, so this selects exactly one shift over any field. For every equation $F(p) = 0$ defining the padded X_j, impose $u_j F(p) = 0$. In selector component j the first block therefore belongs exactly to the padded X_j. Its unused labels are fixed, and its second block is the prescribed shift. Each point of X_j has exactly one image, and restriction to the first block recovers it. This proves the assertion, including preservation of counts. □

Applied to the framed inventories above, the proposition gives a single permutation torsor for all histories within the chosen budget. Forgetting frames is a subsequent symmetry quotient; it is not needed to remove padding multiplicities. This is an encoding of finite histories, not a claim that every rewrite is a permutation of observable states: distinct states can have the same successor. Arbitrarily large histories require a family of budgets. No fixed finite permutation group describes unboundedly many occurrences.

3.2 Schedules change the induced measure

Let Ω be a finite set of acyclic wirings for fixed inventory, and let a finite relabelling group H act on it, preserving the causal relations. Write $\ell(\omega) = |\mathcal{L}(P_\omega)|$. The group acts on scheduled wirings (ω, s) as well.

Proposition 3.3. *The groupoid mass of the scheduled fibre over $[\omega]$ is*

$$\sum_{[(\omega',s)]\mapsto[\omega]} \frac{1}{|\operatorname{Aut}(\omega', s)|} = \frac{\ell(\omega)}{|\operatorname{Aut}(\omega)|}. \tag{3.2}$$

Thus uniform framed scheduled histories induce weights proportional to $\ell(\omega)/|\operatorname{Aut}(\omega)|$ on unlabelled causal histories. If μ_{ord} is this probability, then for any function a on causal history types,

$$\mathbb{E}_{\mu_{\mathrm{causal}}} a = \frac{\mathbb{E}_{\mu_{\mathrm{ord}}}[a/\ell]}{\mathbb{E}_{\mu_{\mathrm{ord}}}[1/\ell]}, \qquad \mu_{\mathrm{causal}}(\omega) \propto \frac{1}{|\operatorname{Aut}(\omega)|}. \tag{3.3}$$

Proof. The stabilizer $\operatorname{Aut}(\omega)$ acts on its $\ell(\omega)$ schedules. Orbit–stabilizer gives (3.2), including any automorphisms of ports or vertices that fix the schedule. For a fixed ambient group H, uniform sampling of framed points has these groupoid weights after forgetting labels. Multiplying by $1/\ell$ and normalizing proves (3.3). □

The linear-extension factor is additional to the incidence symmetry factor. Dividing only by graph automorphisms does not remove bias introduced by choosing sequential schedules. The identity is a finite counting statement; computing ℓ need not be easy.

For a small example, start with two x tokens and use distinguished events $A : x \to x$, $B : x \to y$, $C : x \to z$, $D : y \to w$, with final tokens z, w. There are four compatible acyclic framed wirings. Two put A before B, giving a three-event chain and a singleton, hence four schedules. Two put A before C, giving two two-event chains, hence six schedules. Uniform scheduled histories therefore assign probability $2/5$ to the first type; uniform causal wirings assign $1/2$. Equation (3.3) restores the latter value exactly.

4 The geometry of one occurrence history

Let P be a finite causal poset. A prefix records the events already executed, with their identities retained. Such a set is an order ideal. Conversely every ideal occurs as a prefix, by placing a linear

extension of it before a linear extension of its complement. Write $\mathcal{J}(P)$ for the ideals, and join two ideals when they differ by one event.

Theorem 4.1 (Standard ideal geometry)**.** *The prefix graph is median, with*

$$\begin{aligned} d(J,K) &= |J \triangle K|, \\ m(J,K,L) &= (J \cap K) \cup (K \cap L) \cup (L \cap J). \end{aligned} \tag{4.1}$$

Its cubes, obtained by toggling independent available events, form a contractible CAT(0) cube complex $C(P)$.

Proof. Remove $J \setminus K$ in maximal-first order. An element being removed cannot have a successor in K, since K is an ideal; all its successors outside K have already been removed. Then add $K \setminus J$ in minimal-first order. This gives a path of length $|J \triangle K|$. Every edge changes one coordinate, so no shorter path exists.

The majority in (4.1) is an ideal: every predecessor of an event present in at least two ideals is present in those same two ideals. It is the unique point in all three pairwise distance intervals, as follows coordinatewise from the distance formula.

An explicit cubical realization is

$$C(P) = \{x \in [0,1]^P : x_j > 0 \Longrightarrow x_i = 1 \text{ for every } i < j\}. \tag{4.2}$$

The coordinates strictly between zero and one form an antichain, and all their predecessors are fixed at one. Consequently this is exactly the union of the stated cubes and their faces. If p is maximal in P, multiplying x_p by $1 - t$ gives a deformation retraction to $C(P \setminus \{p\})$. Iteration contracts the complex.

At an ideal J, the available toggles are the maximal elements of J and the minimal elements of $P \setminus J$. A collection of toggles forms a cube exactly when its events are pairwise incomparable. Hence every clique in a vertex link spans a simplex: all links are flag. Contractibility and the cubical link criterion [1, Theorem 2.1] imply CAT(0). □

This proof is included to specify the metric and the information retained. The result is the conflict-free case of established event-structure geometry. It makes no assertion about the graph obtained by identifying prefixes with equal token states.

Proposition 4.2. *The graph of linear extensions, with adjacent incomparable swaps, is geodesically convex in the permutation graph. The distance is the number of event pairs ordered differently.*

Proof. Every adjacent swap changes one pair. To reach a target order, move its first event to the front of the current order, and continue inductively. Each crossed event is incomparable with it, since a comparison would contradict one of the two orders. Each swap fixes one differing pair, attaining the lower bound. Every shortest path in the full permutation graph changes only differing pairs, each once. Comparable pairs agree at both ends, so all intermediate orders remain linear extensions. □

The schedule graph in this proposition and the prefix graph in Theorem 4.1 have different vertices and metrics. Only the latter is the cube complex considered below.

5 Which observations inherit the median?

Let $\sigma : \mathcal{J}(P) \to S$ be any map onto its image. Define $J \sim K$ when $\sigma(J) = \sigma(K)$. We say that the median descends if

$$\sigma(m(J, K, L)) \tag{5.1}$$

depends only on $\sigma(J), \sigma(K), \sigma(L)$. This asks for compatibility with the given history median, a stronger condition than the image graph merely admitting some median operation.

Theorem 5.1 (Observation criterion)**.** *The median descends if and only if there is a subset $Q \subseteq P$ such that*

$$\sigma(J) = \sigma(K) \quad \Longleftrightarrow \quad J \cap Q = K \cap Q. \tag{5.2}$$

The subset is unique. Its complement consists of the events whose edges are collapsed by σ. The quotient prefix graph is the ideal graph of the induced poset $P|_Q$, with its canonical CAT(0) cubical completion.

Proof. Suppose the median descends. Each equivalence class is convex. Indeed, if $J \sim K$ and $L \in [J, K]$, then

$$\sigma(L) = \sigma(m(J, K, L)) = \sigma(m(J, J, L)) = \sigma(J).$$

Now suppose the edge $J, J \cup \{e\}$ is collapsed. For any other edge $K, K \cup \{e\}$ with that event, coordinatewise majority gives

$$m(J, K, K \cup \{e\}) = K, \qquad m(J \cup \{e\}, K, K \cup \{e\}) = K \cup \{e\}.$$

Thus this edge is also collapsed. Collapsing an event is independent of the context in which it is available.

Let D be the set of collapsed events and $Q = P \setminus D$. If $J \sim K$, every shortest path between them lies in their convex class. It crosses each event of $J \triangle K$ once, so these events all belong to D. Hence $J \cap Q = K \cap Q$. Conversely, if these intersections agree, a shortest path changes only events of D. All its edges are collapsed, giving $J \sim K$. Every event has an edge, for example $\downarrow e \setminus \{e\}, \downarrow e$. This also proves uniqueness of Q.

Conversely, restriction to Q commutes with coordinatewise majority. Equation (5.2) therefore makes (5.1) well defined. Every ideal A of $P|_Q$ lifts to $\downarrow_P A$, whose intersection with Q is A. An edge in $\mathcal{J}(P)$ restricts to an edge or a vertex in $\mathcal{J}(P|_Q)$. If $A, A \cup \{q\}$ is an edge in the latter, the ideal

$$J = \downarrow_P A \ \cup \ (\downarrow_P q \setminus \{q\})$$

has visible part A, and $J, J \cup \{q\}$ lifts the edge. Thus the simple quotient graph is precisely $\mathcal{J}(P|_Q)$. Theorem 4.1 identifies its cubical completion. □

The theorem is the ideal-coordinate version of a median quotient obtained by discarding walls. Its elementary proof gives a useful test in terms of observations, without requiring a separate median structure on S in advance.

Corollary 5.2. *The median descends if and only if both conditions hold: every observation fibre is convex, and for each event either all its edges are collapsed or none are.*

Proof. Necessity was proved above. Under these two conditions, the same shortest-path argument proves (5.2). □

Proposition 5.3. *For Q as in Theorem 5.1, the fibre over an ideal A of $P|_Q$ is the convex interval*

$$[\downarrow_P A,\ P \setminus \uparrow_P (Q \setminus A)]. \tag{5.3}$$

Coordinate projection maps $C(P)$ onto $C(P|_Q)$ and every target cube has a cubical lift.

Proof. An ideal with visible part A contains all predecessors of A and excludes all successors of $Q \setminus A$. Conversely every ideal between these bounds has visible part A. The bounds are ideals, and the lower lies in the upper because A is an ideal of the induced poset. Intervals of ideals are convex by (4.1).

Projection of (4.2) satisfies every order constraint on Q. To lift a target cube, let A be its bottom ideal and B its antichain of available events. Set

$$J = \downarrow_P A \ \cup \bigcup_{b \in B} (\downarrow_P b \setminus \{b\}).$$

Its visible part is A, all elements of B are available at J, and no one of them belongs to J. The corresponding cube of $C(P)$ projects onto the given cube. □

Example 5.4. For two independent events e, f, the prefix graph is a square. The observation $\sigma(J) = |J|$ identifies $\{e\}$ and $\{f\}$. Its simple quotient graph is the three-vertex path, which is median. Nevertheless the history median does not descend:

$$m(\{e\}, \{f\}, \varnothing) = \varnothing, \qquad m(\{e\}, \{e\}, \varnothing) = \{e\}.$$

The input observations agree, but the output observations do not. This explains why Theorem 5.1 is not a classification of all state graphs that happen to be median.

6 Token-effect matrices and explicit failure witnesses

Let A be an integer matrix whose column a_e is the change of the recorded token counts under event e. For an ideal J, the observable marking is

$$\sigma_A(J) = b_0 + A\mathbf{1}_J. \tag{6.1}$$

This formula applies to typed-token counts and to any fixed additive observation of a distinguished occurrence history. Nonnegative markings and executability are supplied by the history construction; the algebraic test below needs only (6.1). Let $D = \{e : a_e = 0\}$ and $Q = P \setminus D$. An observation that also forgets vertex names and retains only a hypergraph isomorphism type need not be additive; in that case Theorem 5.1 applies directly.

Theorem 6.1. *The history median descends to the marking observation if and only if*

$$A(\mathbf{1}_J - \mathbf{1}_K) = 0 \quad \Longrightarrow \quad J \triangle K \subseteq D \qquad (J, K \in \mathcal{J}(P)). \tag{6.2}$$

In particular, if every event has nonzero effect, descent is equivalent to injectivity of the marking on ideals. Whenever (6.2) *fails, a shortest prefix path between a colliding pair gives two triples with the same observed inputs and different observed medians.*

Proof. Every edge for event e changes the observation by a_e, independently of context. Hence its edges are collapsed exactly when $e \in D$. Theorem 5.1 now gives exactly (6.2); the converse implication from equality on Q to equality of markings is automatic.

For the witness, choose J, K with equal marking but a nonzero-effect event in their symmetric difference. A shortest path between them changes each differing event once. It therefore contains an edge with nonzero observed change. Some vertex L of this path has $\sigma_A(L) \neq \sigma_A(J)$. Since $L \in [J, K]$,

$$m(J, K, L) = L, \qquad m(J, J, L) = J.$$

The two triples have the same observed inputs and different observed medians, as required. □

This test concerns only differences of ideal indicators, not the whole integer kernel of A. Requiring $\ker A = 0$ would be unnecessarily strong. For a chain $e < f$, columns $a_e = 1$, $a_f = 2$ have a nontrivial integer relation, but the ideal markings $0, 1, 3$ are distinct and the history median descends.

The criterion also gives a minimal repair of an additive observation. For every colliding pair that differs on a nonzero-effect event, retain at least one differing event coordinate. This is a finite separation problem, not automatically a small one.

Proposition 6.2. *For $R \subseteq P$, augment the marking by $J \cap R$. Its median descends if and only if R meets every set*

$$J \triangle K \quad \text{with} \quad A\mathbf{1}_J = A\mathbf{1}_K, \qquad (J \triangle K) \cap Q \neq \varnothing. \tag{6.3}$$

Thus the least number of additional event bits is the transversal number of this explicitly defined collision family.

Proof. The augmented effect is nonzero exactly on $Q \cup R$. Two augmented observations agree precisely when the markings agree and $J \cap R = K \cap R$. Under that agreement, their difference contains no event of R, so it meets $Q \cup R$ exactly when it meets Q. Theorem 6.1 proves the equivalence. □

If only bits from Q may be retained, replace each set in (6.3) by its intersection with Q.

For two independent events of equal effect, the collision family contains $\{e, f\}$ and one event bit repairs the observation. More generally, for n independent events with the same nonzero effect, the minimum is $n - 1$: every pair $\{e, f\}$ occurs as a collision, so at most one coordinate may be omitted. Conversely, the total count and any $n - 1$ bits determine the final bit.

7 Confluence permits arbitrary free rank

A state complex has one edge for each rewrite occurrence at a state and cubes for independent simultaneously executable rewrites. A diamond caused by alternative competing rules does not by itself provide a commuting square. This distinction is standard in state complexes and Petri-net cube complexes [5, 4].

Theorem 7.1. *For every $k \geq 2$ and $L \geq 3$ there is a finite graded, terminating, confluent one-token rewriting system such that:*

1. *its state complex has dimension one, girth $2L$, and fundamental group F_{k-1};*
2. *each complete occurrence history has prefix complex an interval of L edges;*
3. *its full history unfolding from the initial state is a tree.*

Its simple state graph is not median, and its state complex is not CAT(0).

Proof. Take states s, t and $u_{j,a}$ for $1 \le j \le k$, $1 \le a < L$. The only rules are the k chains

$$s \longrightarrow u_{j,1} \longrightarrow \cdots \longrightarrow u_{j,L-1} \longrightarrow t. \tag{7.1}$$

Give these states levels $0, a, L$ respectively. Every rule increases the level by one, proving termination. Every reachable state reduces to t, so any two reducts have the common reduct t; this proves confluence. Each rule consumes the unique token, hence no two occurrences are independent and there are no higher-dimensional cubes.

The underlying graph has $k(L-1)+2$ vertices and kL edges. It is connected, so its first Betti number is $k-1$ and its fundamental group is F_{k-1}, by collapsing a spanning tree. Every simple cycle uses two entire branches, giving girth $2L$. A complete history chooses one branch and has a chain of L events. Its ideal complex is an interval. The full unfolding retains the k final occurrences separately; it consists of the k branches joined only at the root, and is a tree.

To see that the state graph is not median, use the vertices $s, u_{1,L-1}, u_{2,2}$. Their three pairwise shortest paths have lengths $L-1, L-1, 2$, all strictly less than L. They are the three consecutive arcs of the two-branch cycle and have no common vertex. An additional branch cannot shorten any of them, since traversing it already costs L. Finally a one-dimensional CAT(0) complex is a tree, whereas this complex has nontrivial fundamental group. □

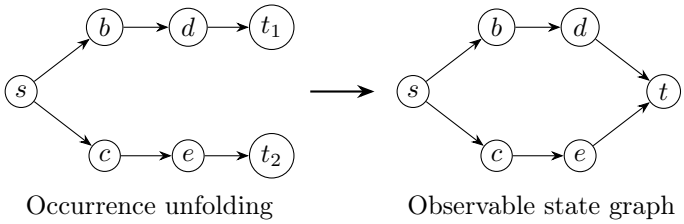


Figure 1: The case $k = 2$, $L = 3$. Identifying the final occurrences creates a six-cycle. No independent events supply a square.

These one-token rules can be read as replacements of typed unary hyperedges at one immutable vertex. They are therefore instances of the model in Theorem 3.1. The example concerns a restricted family of rewrite systems; it does not say that every state quotient has nonpositive curvature or that every system with confluence fails to preserve geometry.

8 Scope and reproduction

There are three separate operations. Relabelling incidence positions gives a symmetry quotient with inverse-automorphism weights. Forgetting schedules gives linear-extension fibres and the extra weight in (3.2). Observing token states makes further identifications, whose compatibility with the history median is tested by Theorems 5.1 and 6.1. Neither of the first two operations proves that the last preserves geometry.

The known CAT(0) description supplies an exact finite geometry of occurrences. A geometric assertion about observable states needs control of the observation map. The repair problem in Proposition 6.2 makes that requirement quantitative: one must retain enough event information to separate the relevant collisions. A continuum limit or a physical field equation would require additional hypotheses and estimates, none of which is implied by the torsor presentation.

The exact checker `check_history_states.py` compares wiring restrictions with direct sequential execution, checks the incidence weights, and exhaustively tests the observation criterion on all partitions of ideal sets for naturally labelled posets through three events. It also checks coordinate quotients, additive observations through four events, repair sets, and the confluent families above. The finite calculations supplement the proofs. Run `make history-state-check` from the repository root. All source inputs are TeX and bibliography files; the figure is drawn directly in TeX. The recorded run checks 4 890 partitions, of which 66 inherit the median, 706 coordinate quotients and cube lifts, 3 450 additive observations and 2 230 repair sets. It constructs 2 742 explicit failure witnesses and checks 16 confluent systems of the form (7.1).

References

[1] F. Ardila, M. Owen, and S. Sullivant. Geodesics in CAT(0) cubical complexes, 2012. URL: `https://arxiv.org/abs/1101.2428`, `arXiv:1101.2428`.

[2] J. C. Baez, A. E. Hoffnung, and C. D. Walker. Higher-dimensional algebra VII: Groupoidification, 2009. URL: `https://arxiv.org/abs/0908.4305`, `arXiv:0908.4305`.

[3] B. H. Bowditch. Median algebras, 2024. Revised 12 May 2024. URL: `https://warwick.ac.uk/fac/sci/maths/people/staff/brian_bowditch/median-algebras.pdf`.

[4] J. Chalopin and V. Chepoi. 1-safe Petri nets and special cube complexes: Equivalence and applications, 2019. URL: `https://arxiv.org/abs/1810.03395`, `arXiv:1810.03395`.

[5] R. Ghrist and V. Peterson. The geometry and topology of reconfiguration. *Advances in Applied Mathematics*, 38:302–323, 2007. URL: `https://www2.math.upenn.edu/~ghrist/preprints/statecpx.pdf`, `doi:10.1016/j.aam.2005.08.009`.

[6] A. Kolpakov. The local Whitehead-move chain on cubic multigraphs, 2026. Companion manuscript. URL: `https://github.com/WolframInstitute/graph_of_graphs/blob/main/rule-a-paper/local-whitehead-chain.tex`.

[7] M. Nielsen, G. D. Plotkin, and G. Winskel. Petri nets, event structures and domains, part I. *Theoretical Computer Science*, 13(1):85–108, 1981. `doi:10.1016/0304-3975(81)90112-2`.